# CS-433 Project 1

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## I. Introduction

Cardiovascular Disease (CVD) is a class of disease that affects the heart and/or blood vessels, which may be caused by a combination of many socioeconomic, behavioral, and environmental risk factors. The rise and instauration of new technologies lead to a global increase in life expectancy, but also to an increase in CVDs worldwide, which are rapidly becoming one of the leading causes of death according to the World Health Organization (WHO). Unfortunately, CVD prediction remains difficult because of the highly complex and non-linear relationship between the risk factors known to spur them and the biological responses that may be generated by patients.

Machine learning (ML) algorithms have been found to be a secure and relatively reliable technique for early detection and prevention of CVDs because they can rapidly process and discern complex, high-dimensional patterns in given biological datasets that traditional methods might not detect (quickly enough). This paper aims to display how such ML can help predict coronary heart disease (MICHD) in patients through the step-by-step building and fine-tuning of a classification algorithm based on a dataset extracted from the US Behavioral Risk Factor Surveillance System (BRFSS).

## II. Base Model

Minimal dataset preprocessing was performed at first. Missing values were simply imputed with the variable's median, and the data was standardized. We performed a 5-fold cross-validation. The decision threshold was tuned per fold to maximize the harmonic mean of precision and recall (F1 score).

Basic ML algorithms were implemented and tested: 1) Mean squared error with gradient descent (GD), 2) Mean squared error with stochastic gradient descent (SGD), 3) Least squares regression with normal equations (LS), 4) Ridge regression with normal equations (RR), 5) Logistic regression with gradient descent (LGD), 6) Regularised Logistic regression with gradient descent (RLGD).

Performance metrics for all the base models can be seen in Figure 1. Looking at the $F_1$ metric, we conclude the LS model gives the best performance, closely followed by GD and RR. We can infer these last two methods were under-trained since their F1 metrics during training and validation are approximately identical. The SGD method failed ($AUC < 0.5$), yielding worse results than random classification.

## III. Improvements

The training and validation F1 were nearly equal using this model, suggesting that the model is not overfitting. Because regularization improves model performance by reducing overfitting, it is unlikely that increasing regularization would drastically improve the performance of the model. Improving the information available to the model through additional pre-processing steps is more likely to improve its performance.

### A. One-hot encoding

The first improvement made was one-hot encoding. Many variables contain categorical data stored under numeric code. For example, the `marital_status` can be encoded by `0` for the category `single`, `1` for `divorced`, or `2` for `married`. Such categorical encoding suggests to a linear model that there is a mathematical ordering between categories. In the above case, the model assumes that `divorced` is somewhere between `single` and `married`. However, this is an incorrect assumption. One-hot encoding solves this problem. The idea is to create separate variables for each category, effectively converting the categorical encoding into a binary encoding. Using the same example as before, the variable `marital_status` is transformed into three variables `marital_status_single`, `marital_status_divorced`, and `marital_status_married`. These new variables take the values `0` and `1`, where `1` denotes that the data point belongs to the associated category.

A variable was denoted as categorical if it was integer-valued and it contained at most 10 levels, that is, 10 distinct integer values. Using one-hot encoding, 219 categorical variables were expanded into 659 new variables. Specifically,

i) The most frequent level was dropped to reduce redundancy. A dropped category can be inferred from the remaining categories.

ii) Levels occurring in under 0.05% of samples were grouped into a single `other` level. Such rare categories provide too little observations for the model to learn from.

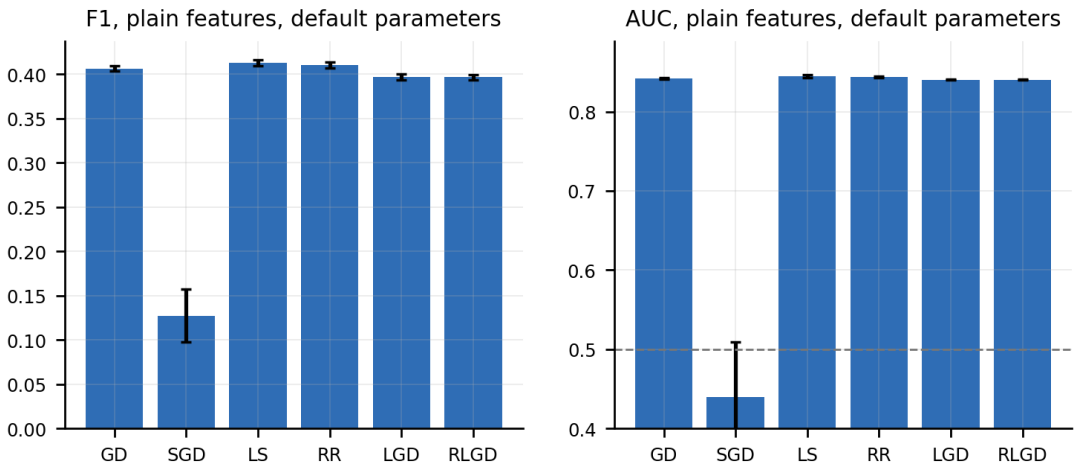



Figure 1. **F1 and AUC metrics for the six base methods.** F1 is the main competition metric and represents the harmonic mean of precision of the method. AUC represent the probability a randomly selected sick patient scores higher than a randomly selected healthy one, independently of the score cut-off above which a person is predicted to have the disease.

iii) Missing values were treated as a distinct level. This allows the model to learn whether a missing value potentially carries information.
This brought the variable count to 782 and improved the model's F1 from 0.412 to 0.430 using Ridge regression.

### *B. Interactions*

The second improvement made was to consider new features to account for interactions between variables. Conditions such as high blood pressure, cholesterol, and diabetes may occur together and their co-occurrence may increase the chances of CVD. Similarly, the effects of having these conditions on CVD may differ according to age or sex. The idea is to create a new variable from a variable pair by multiplying their values, effectively capturing their interactions. For example, `sex` and `cholesterol` may be used to create the variable `sex*cholesterol`, which takes the value of `cholesterol` whenever `sex` belongs to the category `male` and `0` whenever `sex` belongs to the category `female`.
Specifically, we considered interactions between either 2 one-hot encoded variables or 1 one-hot encoded variable and one original variable. We handpicked 19 interactions based on prior domain knowledge about CVD risk factors. These 19 variable pairs were expanded into 96 new interaction variables. This brought the variable count to 878 and improved the model's F1 from 0.430 to 0.437 using Ridge regression.

## IV. Method of Elimination

The final model was arrived at by a method of elimination, which was carried out from the base model through all subsequent improvements.

After training the base models, it was clear that SGD was unstable (very high threshold) and gave poor performance ($AUC < 0.5$). A parametric analysis was done by varying the maximum allowed iterations (`max_iters`), increasing which resulted in slightly better performance from SGD. But for good performance, a lot more iterations would have been required, at which point SGD would effectively become gradient descent. For these reasons, SGD was rejected.

We also realised that a lot of features were correlated, which meant that the matrix $X^TX$ would be singular. With so many features, it's computationally expensive to ensure none are correlated if we decide to drop some. This made LS not an ideal choice since its closed-form solution requires $X^TX$ to be invertible.

This left us with ridge regression, gradient descent, and logistic regression. RR was ideal amongst these to test our improvements on since it has a closed-form solution and would be faster to train and test.

After finalizing all the feature improvements we had to make, we trained all the models still not eliminated with these improvements. Out of these, regularized logistic regression gave the best performance.

## V. Final Model

Improvements on F1 due to the addition of features reported above were computed only on a model using Ridge regression for the sake of time. A new model using regularized logistic regression with a step size $\gamma = 0.01$ and a regularization constant $\lambda = 0.001$ was trained on the new feature set. This improved the model's F1 to 0.439, against the previously computed 0.437.